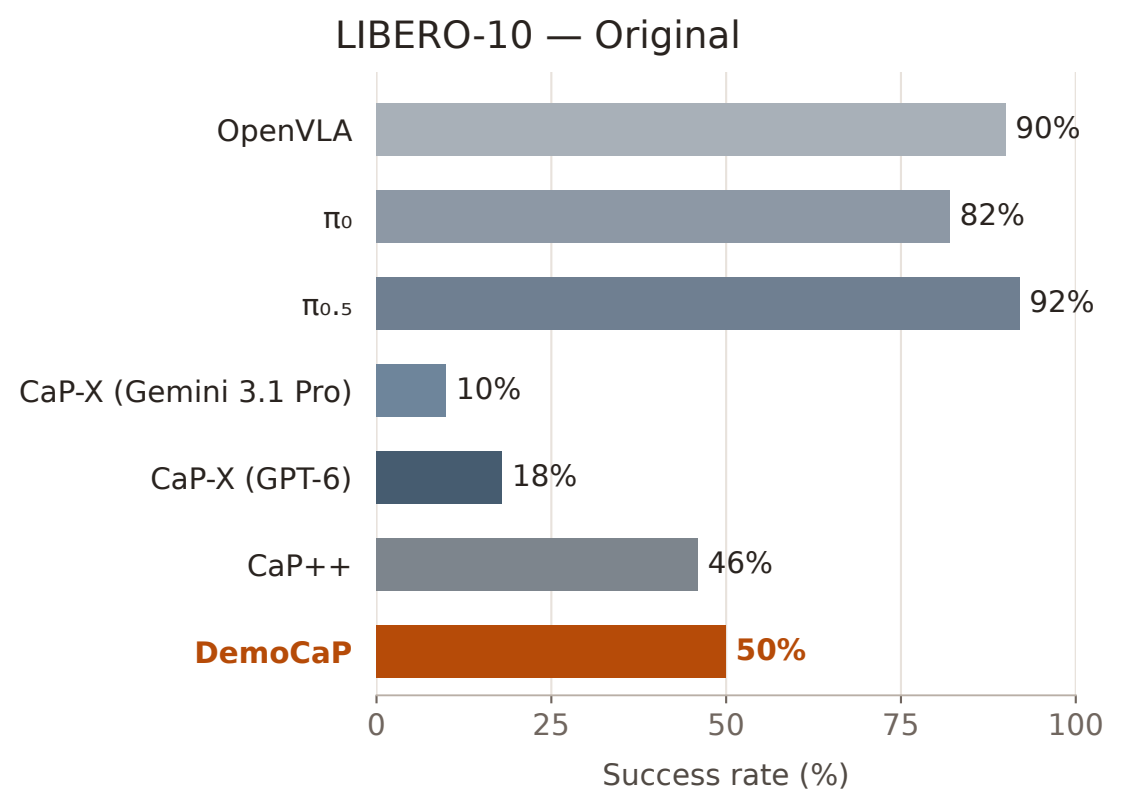
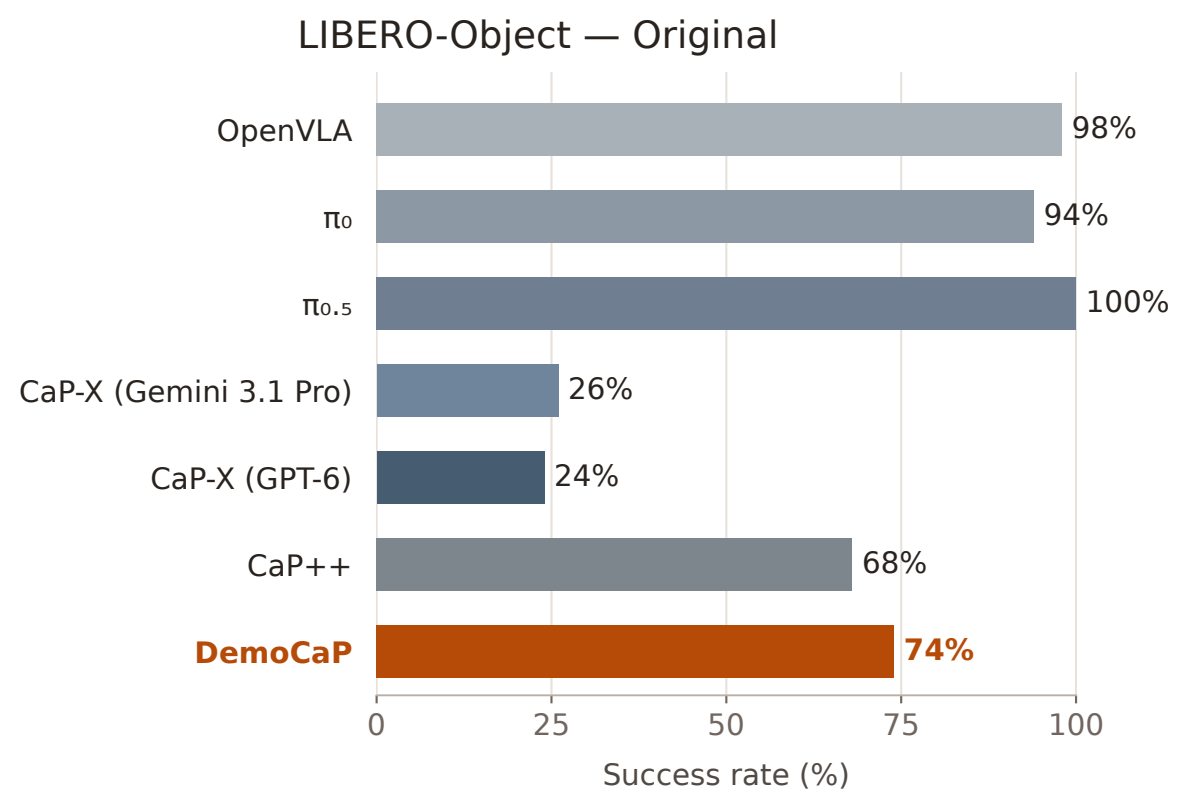
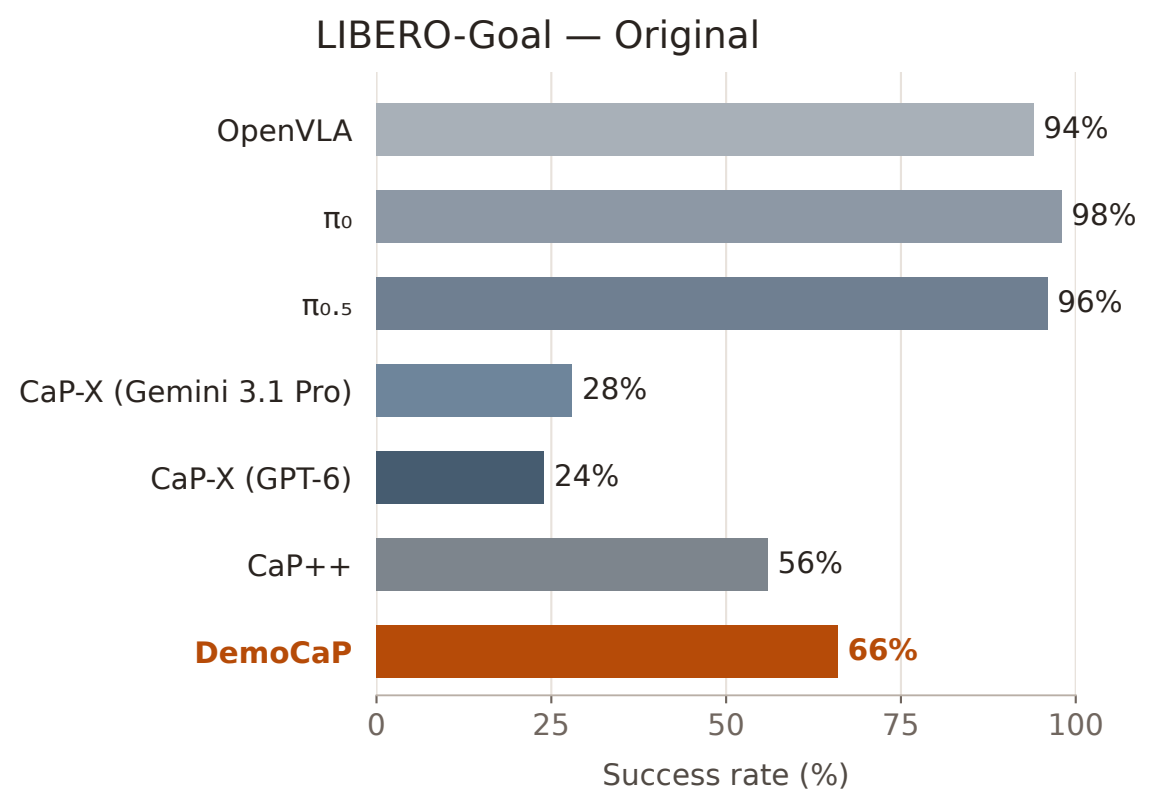
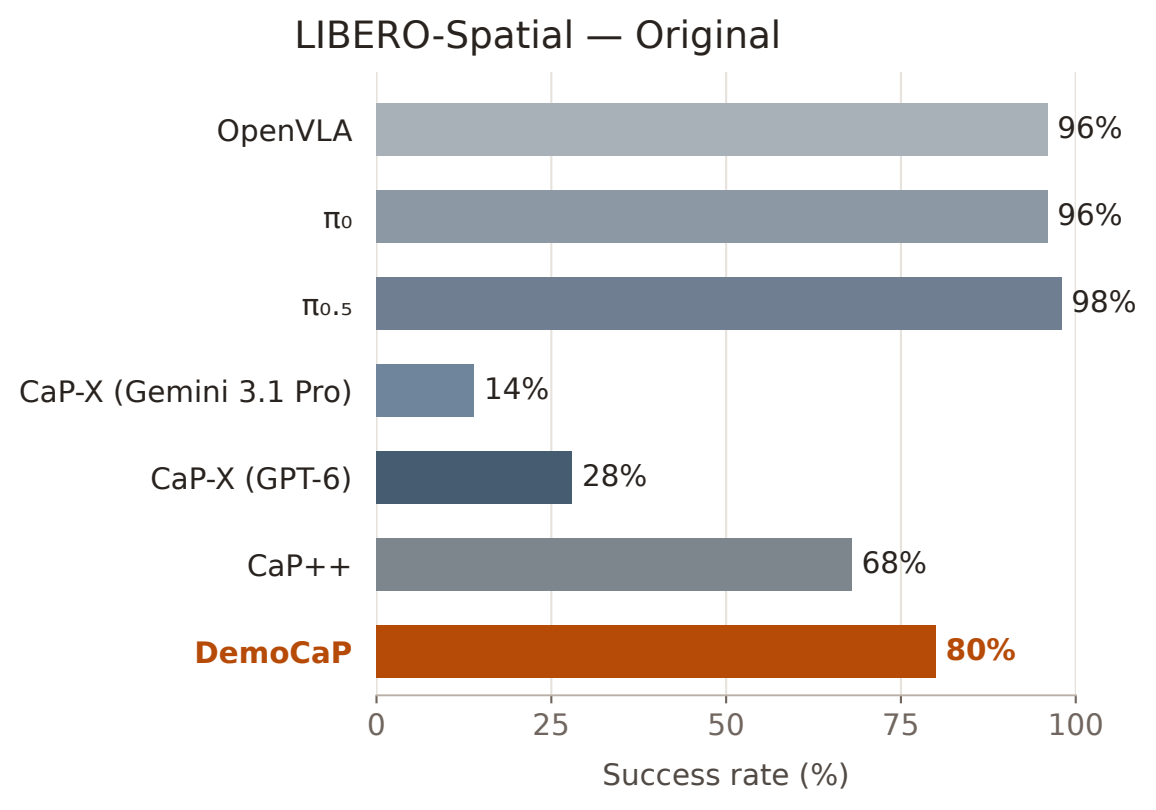
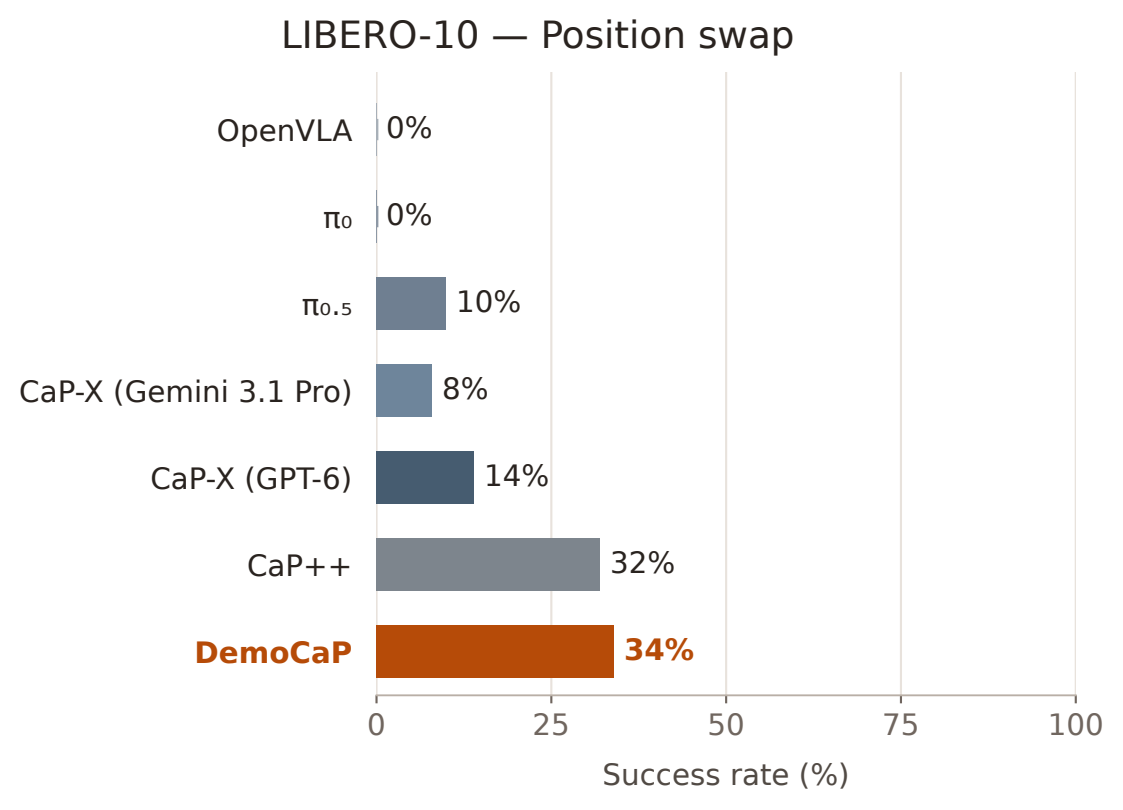
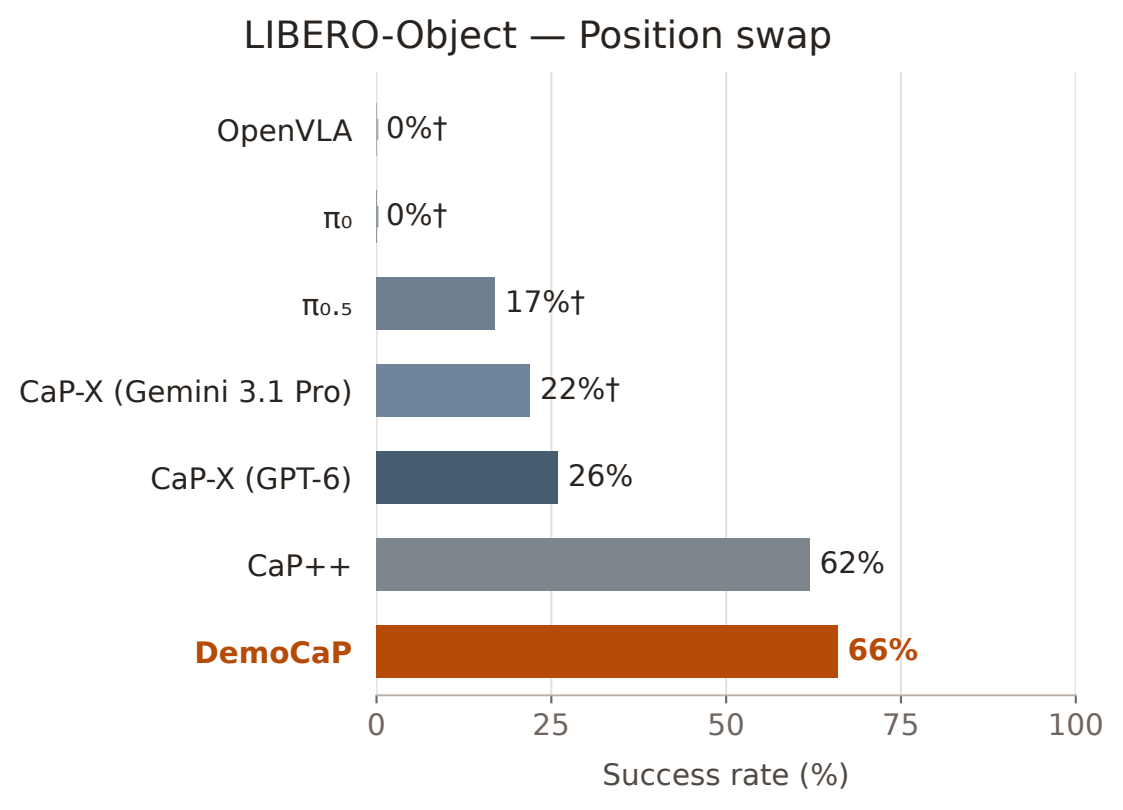

LIBERO-10 — Original
OpenVLA
π0
π0.5
CaP-X (Gemini 3.1 Pro)
CaP-X (GPT-6)
CaP++
DemoCaP
90%
82%
92%
10%
18%
46%
50%
0
25
50
75
100
Success rate (%)

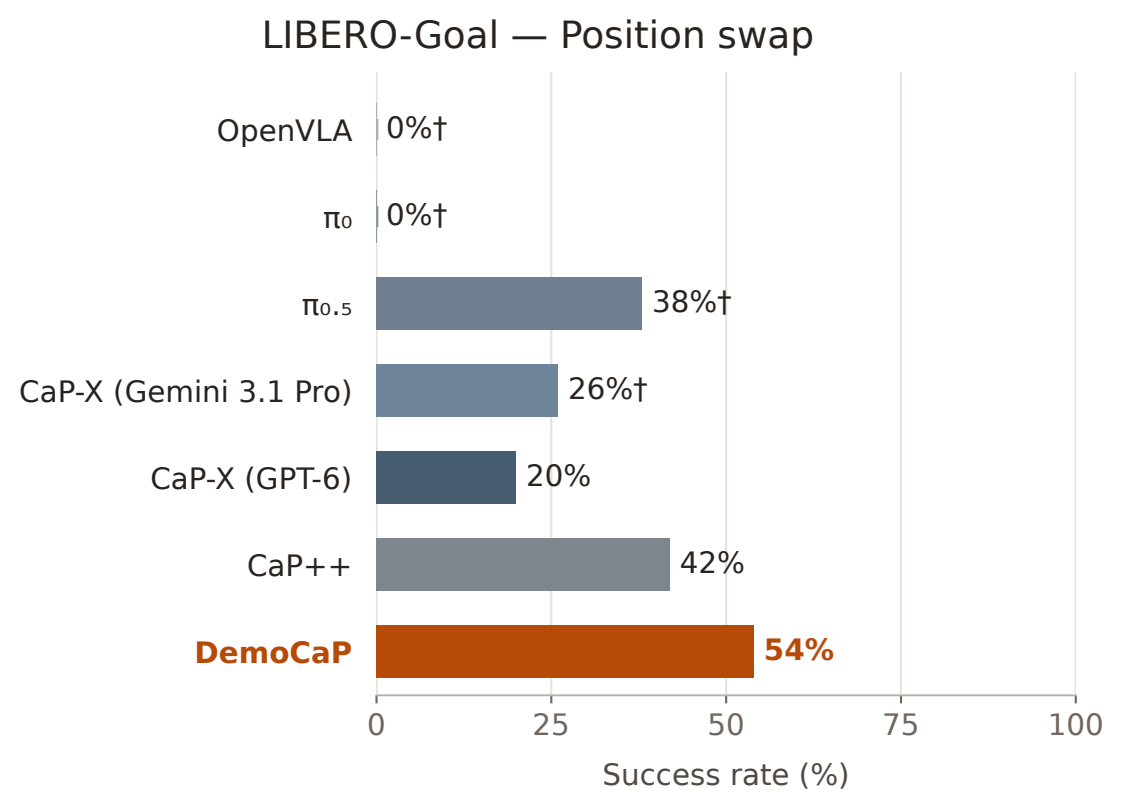

LIBERO-Object — Original
OpenVLA
π0
π0.5
CaP-X (Gemini 3.1 Pro)
CaP-X (GPT-6)
CaP++
DemoCaP
98%
94%
100%
26%
24%
68%
74%
0
25
50
75
100
Success rate (%)

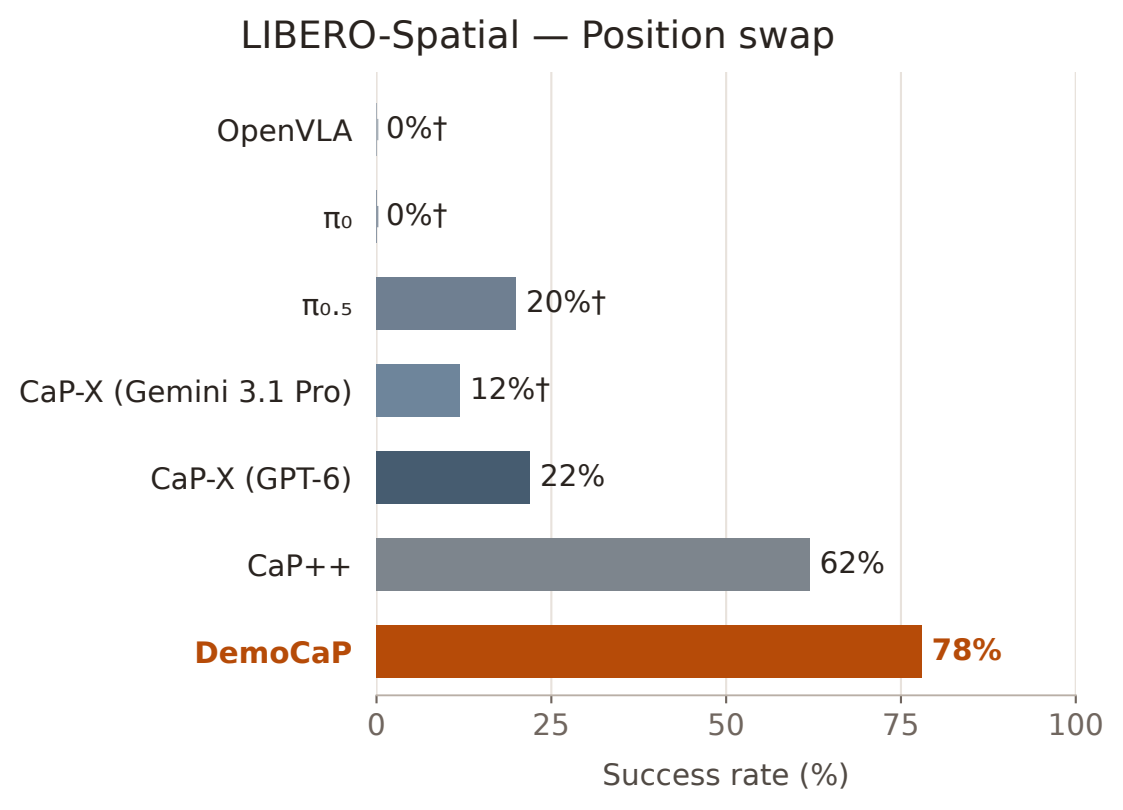

LIBERO-Goal — Original
OpenVLA
π0
π0.5
CaP-X (Gemini 3.1 Pro)
CaP-X (GPT-6)
CaP++
DemoCaP
94%
98%
96%
28%
24%
56%
66%
0
25
50
75
100
Success rate (%)

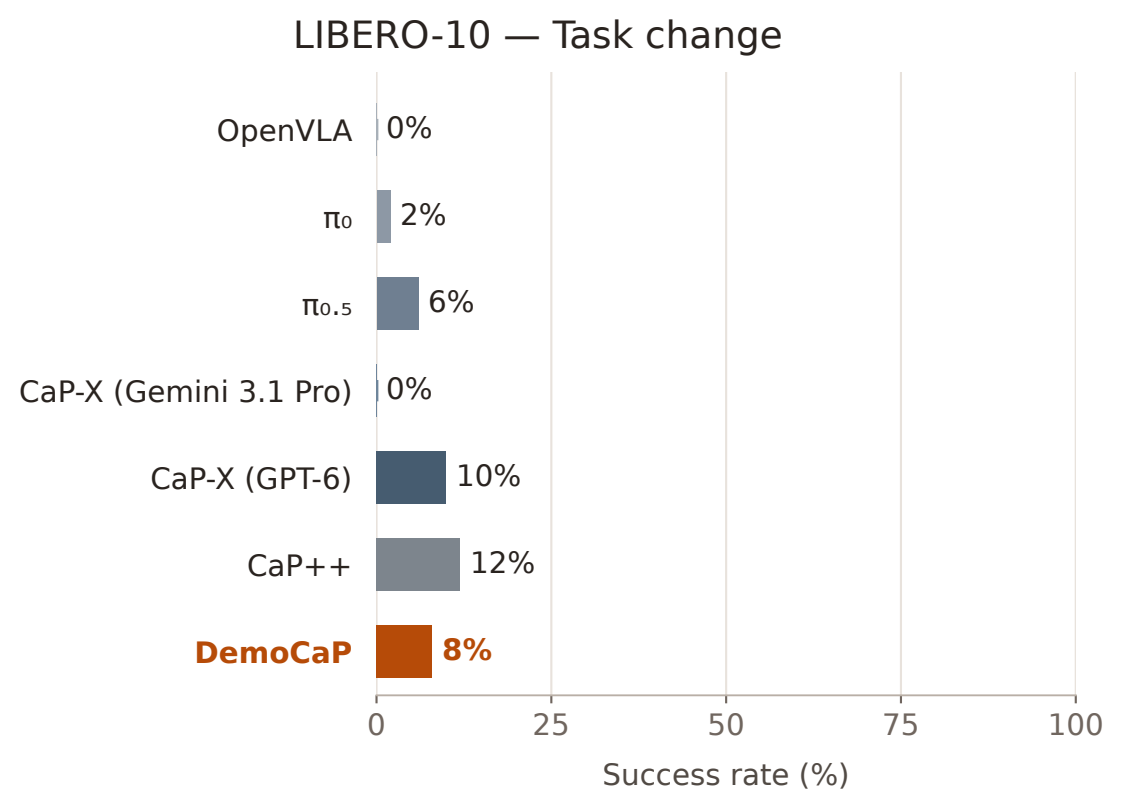

LIBERO-Spatial — Original
OpenVLA
π0
π0.5
CaP-X (Gemini 3.1 Pro)
CaP-X (GPT-6)
CaP++
DemoCaP
96%
96%
98%
14%
28%
68%
80%
0
25
50
75
100
Success rate (%)

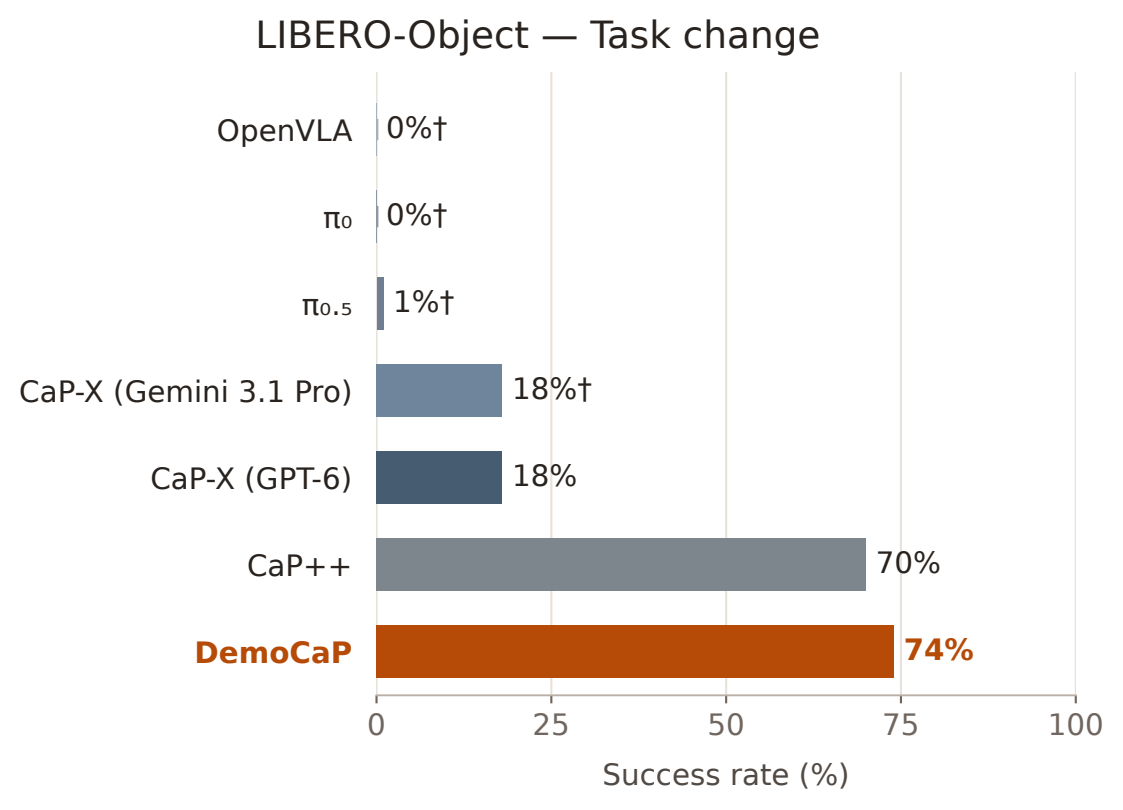

LIBERO-10 — Position swap
OpenVLA
0%
π0
0%
π0.5
10%
CaP-X (Gemini 3.1 Pro)
8%
CaP-X (GPT-6)
14%
CaP++
32%
DemoCaP
34%
0
25
50
75
100
Success rate (%)

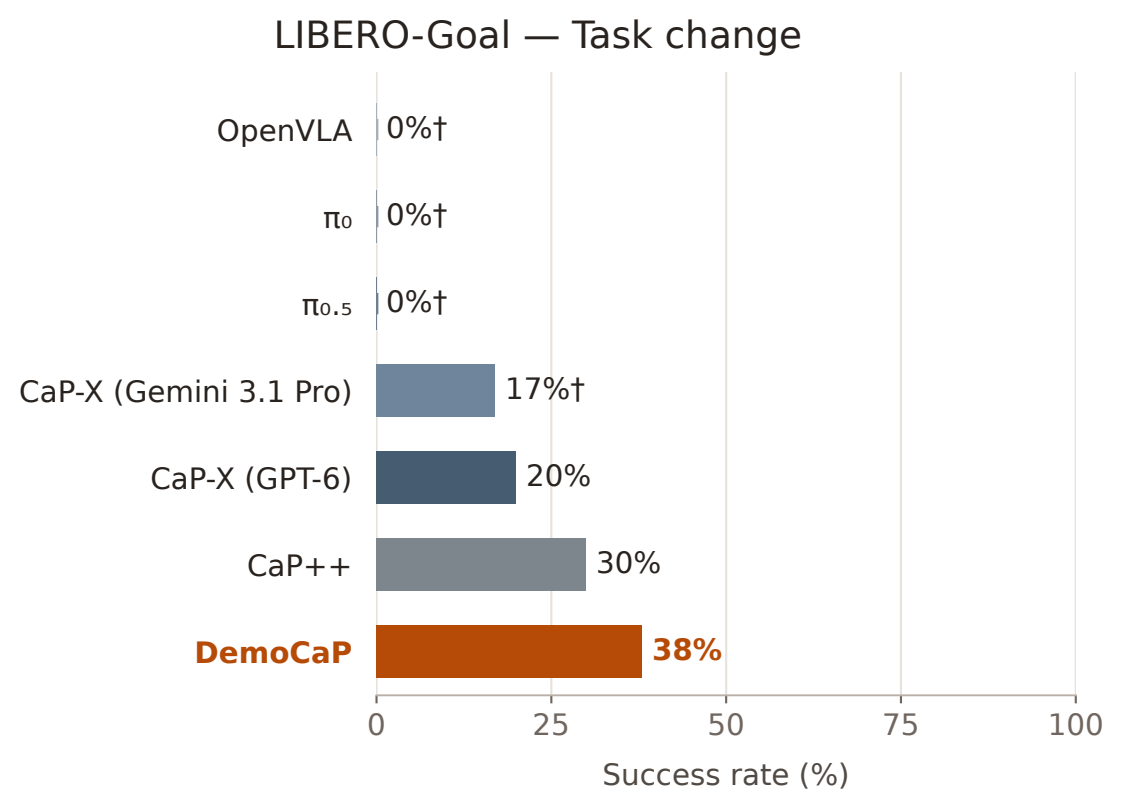

LIBERO-Object — Position swap
OpenVLA
0%†
π0
0%†
π0.5
17%†
CaP-X (Gemini 3.1 Pro)
22%†
CaP-X (GPT-6)
26%
CaP++
62%
DemoCaP
66%
0
25
50
75
100
Success rate (%)

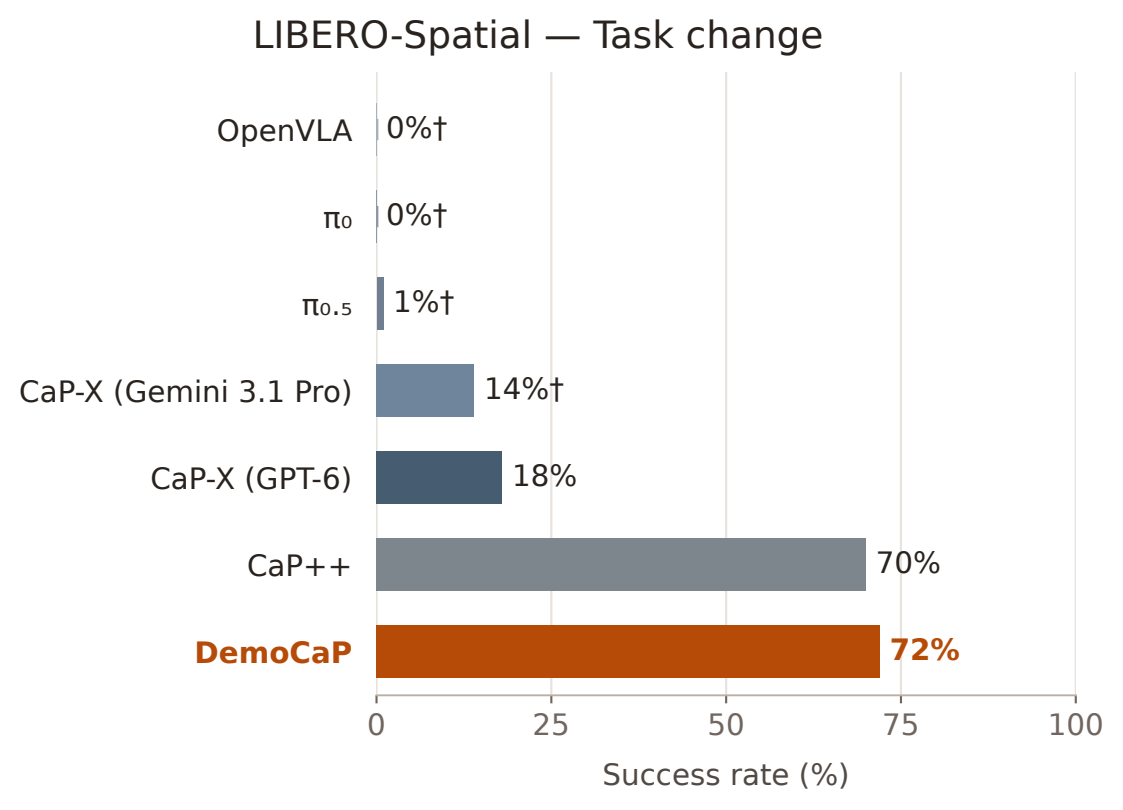

LIBERO-Goal — Position swap
OpenVLA
0%†
π0
0%†
π0.5
38%†
CaP-X (Gemini 3.1 Pro)
26%†
CaP-X (GPT-6)
20%
CaP++
42%
DemoCaP
54%
0
25
50
75
100
Success rate (%)

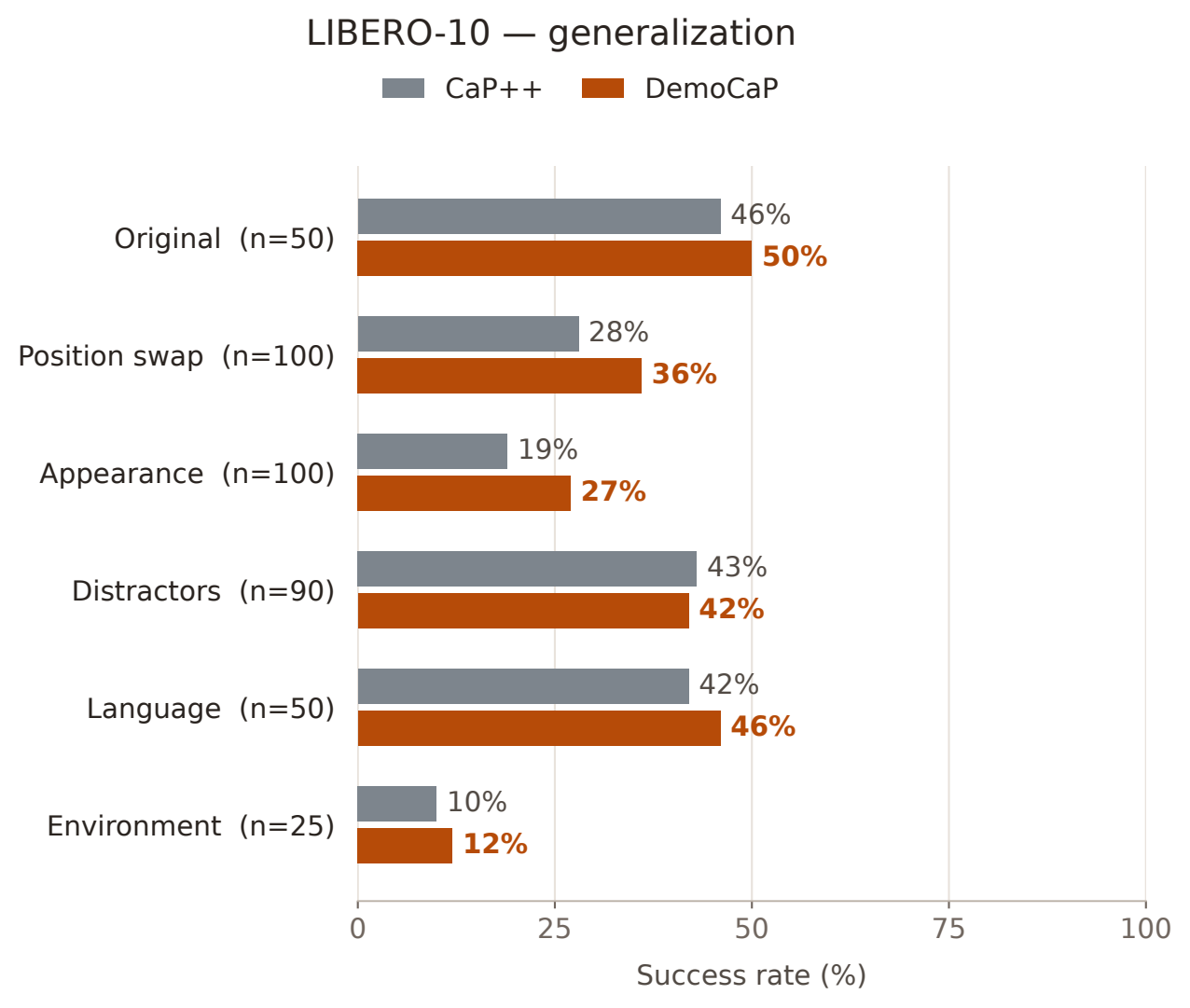

LIBERO-Spatial — Position swap
OpenVLA
0%†
π0
0%†
π0.5
20%†
CaP-X (Gemini 3.1 Pro)
12%†
CaP-X (GPT-6)
22%
CaP++
62%
DemoCaP
78%
0
25
50
75
100
Success rate (%)

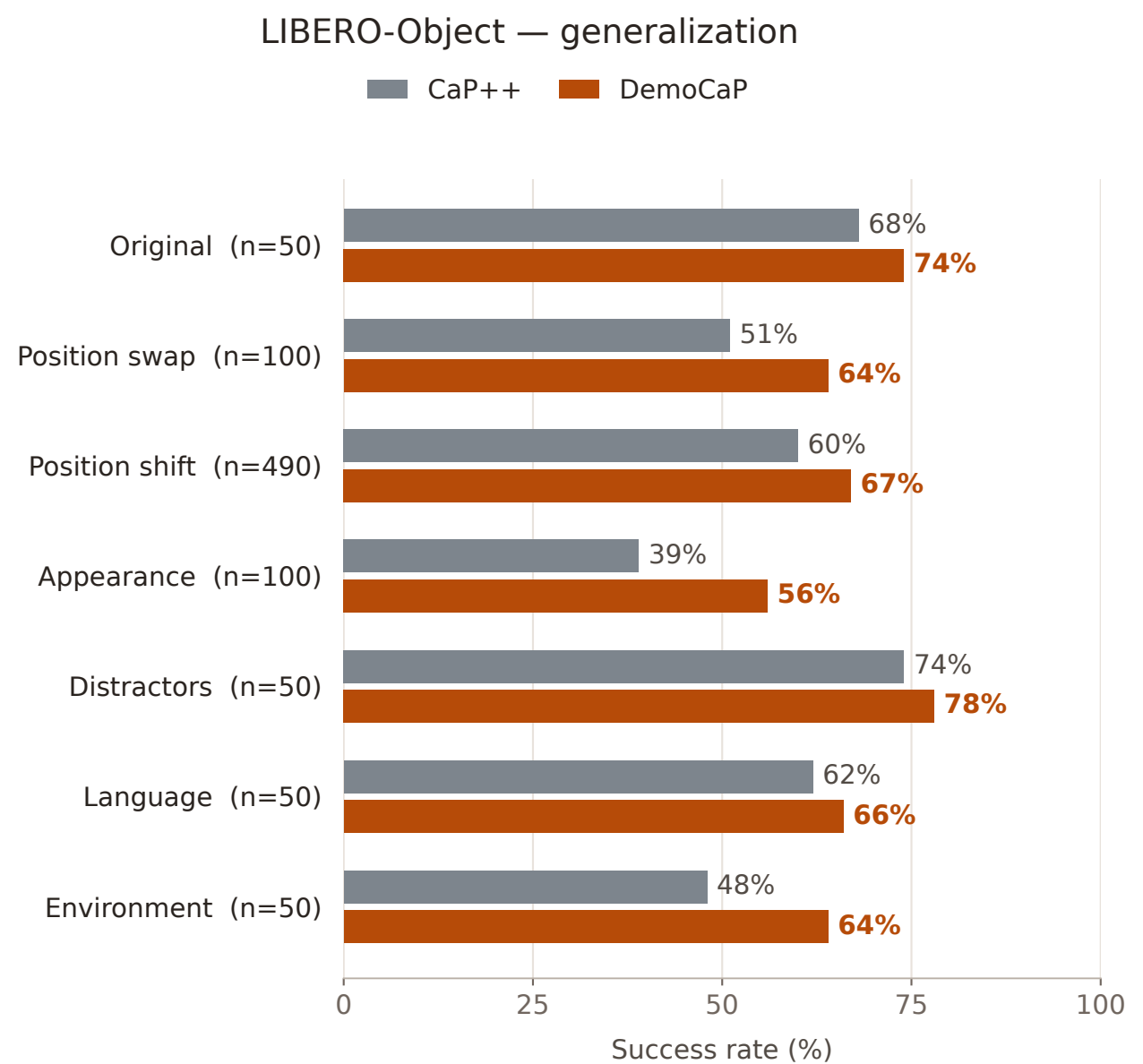

LIBERO-10 — Task change
OpenVLA
0%
π0
2%
π0.5
6%
CaP-X (Gemini 3.1 Pro)
0%
CaP-X (GPT-6)
10%
CaP++
12%
DemoCaP
8%
0
25
50
75
100
Success rate (%)

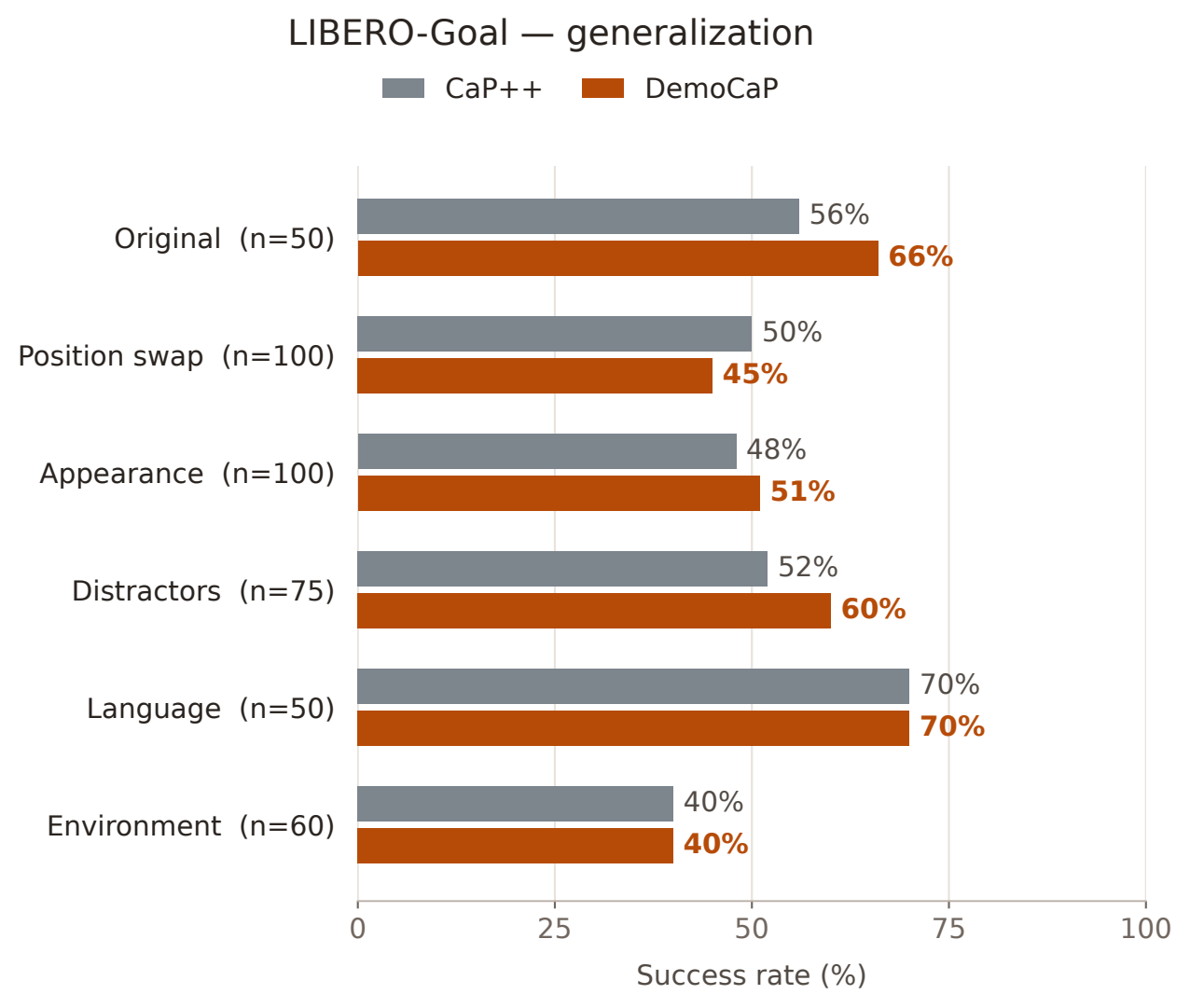

LIBERO-Object — Task change
OpenVLA
0%†
π0
0%†
π0.5
1%†
CaP-X (Gemini 3.1 Pro)
18%†
CaP-X (GPT-6)
18%
CaP++
70%
DemoCaP
74%
0
25
50
75
100
Success rate (%)

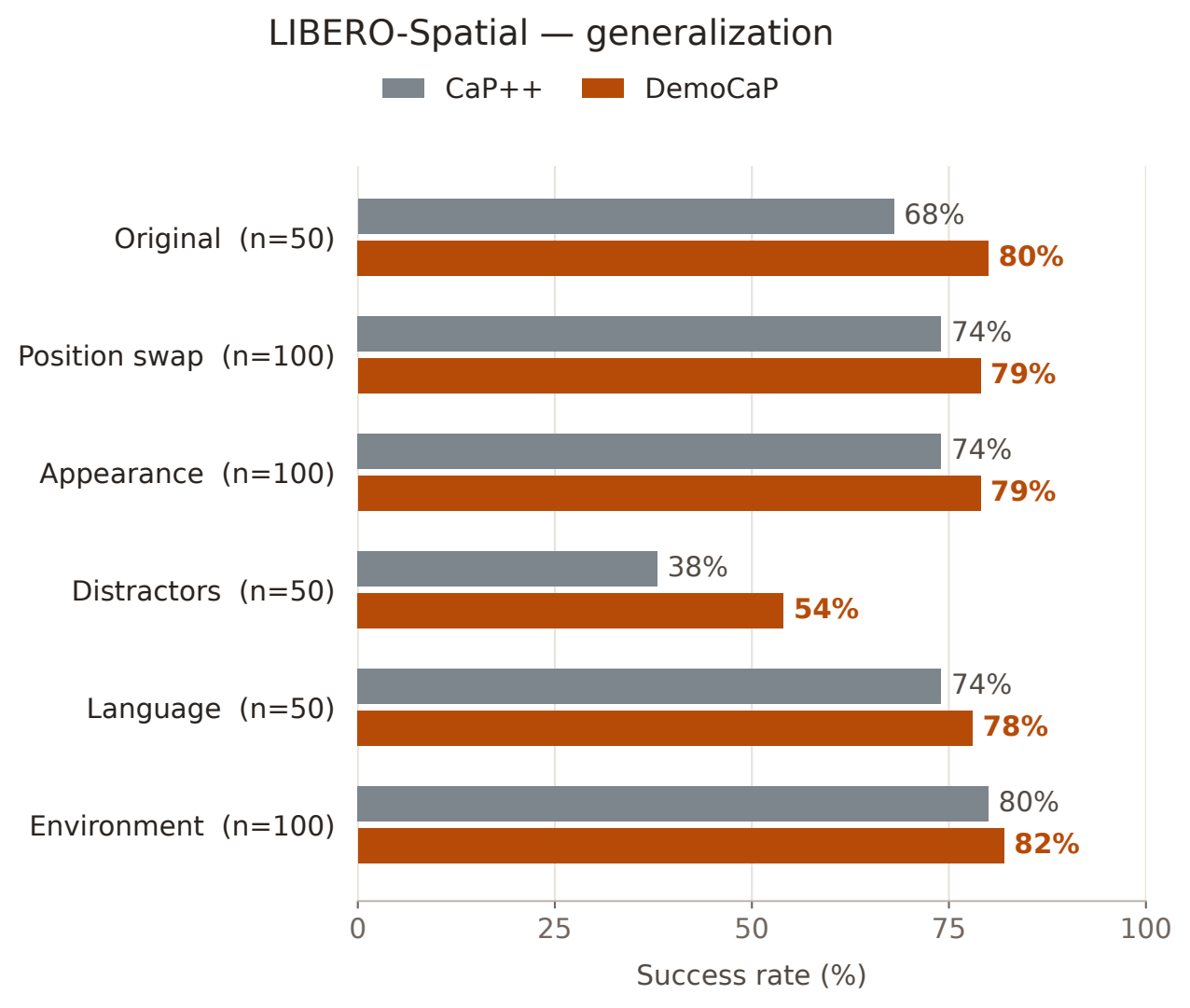

LIBERO-Goal — Task change
OpenVLA
0%†
π0
0%†
π0.5
0%†
CaP-X (Gemini 3.1 Pro)
17%†
CaP-X (GPT-6)
20%
CaP++
30%
DemoCaP
38%
0
25
50
75
100
Success rate (%)

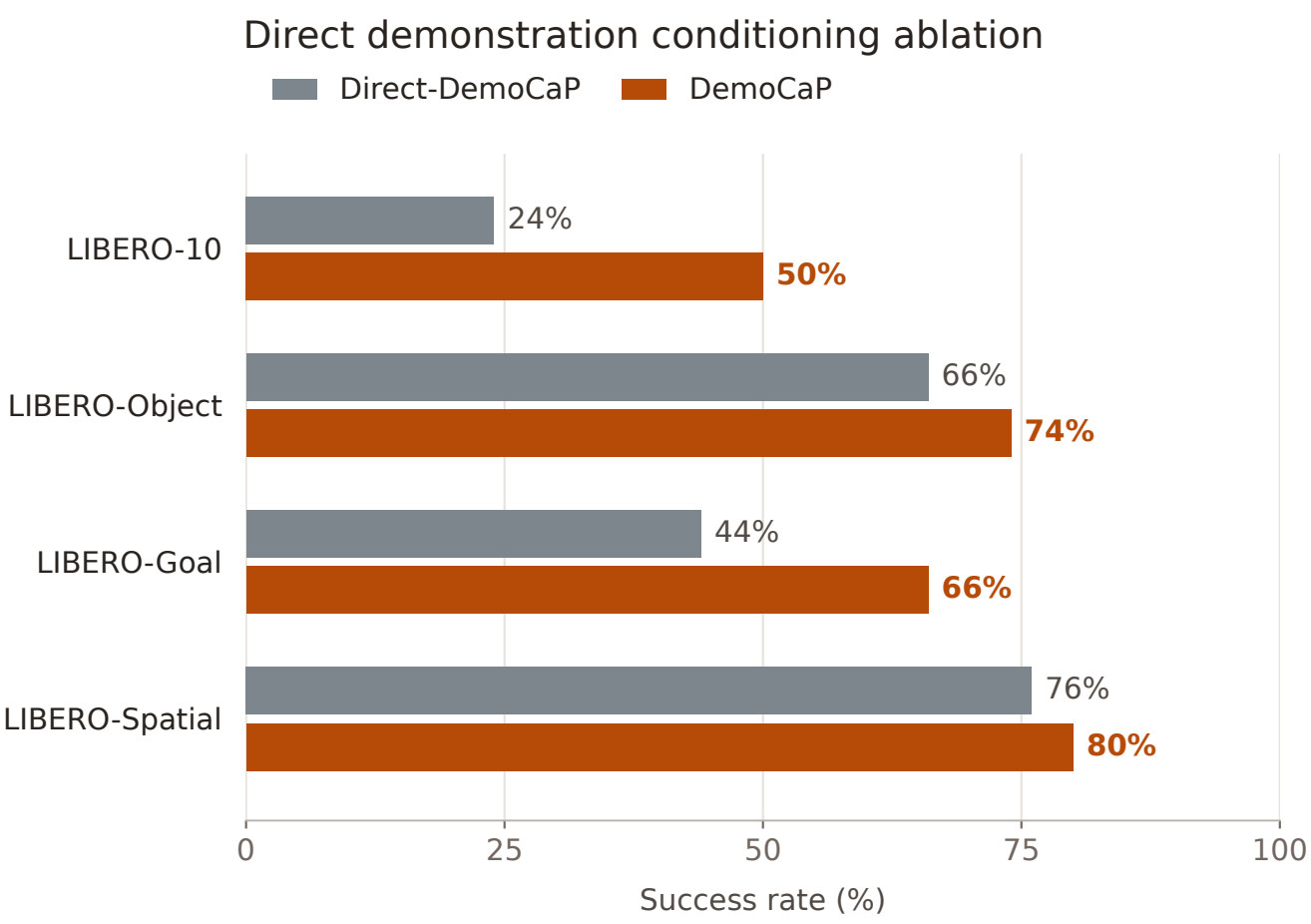

LIBERO-Spatial — Task change
OpenVLA
0%†
π0
0%†
π0.5
1%†
CaP-X (Gemini 3.1 Pro)
14%†
CaP-X (GPT-6)
18%
CaP++
70%
DemoCaP
72%
0
25
50
75
100
Success rate (%)


LIBERO-10 — generalization
CaP++
DemoCaP
Original (n=50)
46%
50%
Position swap (n=100)
28%
36%
Appearance (n=100)
19%
27%
Distractors (n=90)
43%
42%
Language (n=50)
42%
46%
Environment (n=25)
10%
12%
0
25
50
75
100
Success rate (%)


LIBERO-Object — generalization
CaP++
DemoCaP
Original (n=50)
68%
74%
Position swap (n=100)
51%
64%
Position shift (n=490)
60%
67%
Appearance (n=100)
39%
56%
Distractors (n=50)
74%
78%
Language (n=50)
62%
66%
Environment (n=50)
48%
64%
0
25
50
75
100
Success rate (%)


LIBERO-Goal — generalization
CaP++
DemoCaP
Original (n=50)
56%
66%
Position swap (n=100)
50%
45%
Appearance (n=100)
48%
51%
Distractors (n=75)
52%
60%
Language (n=50)
70%
70%
Environment (n=60)
40%
40%
0
25
50
75
100
Success rate (%)


LIBERO-Spatial — generalization
CaP++
DemoCaP
Original (n=50)
Position swap (n=100)
Appearance (n=100)
Distractors (n=50)
Language (n=50)
Environment (n=100)
68%
80%
74%
79%
74%
79%
38%
54%
74%
78%
80%
82%
0
25
50
75
100
Success rate (%)


Direct demonstration conditioning ablation
Direct-DemoCaP
DemoCaP
LIBERO-10
24%
50%
LIBERO-Object
66%
74%
LIBERO-Goal
44%
66%
LIBERO-Spatial
76%
80%
0
25
50
75
100
Success rate (%)